

R CODE LAB EXPERIMENT

SET 8 – Employee Performance Analysis

```
employee <- 1:3
```

```
dept <- c("Sales","HR","Marketing")
```

```
service <- c(5,3,7)
```

```
score <- c(85,92,78)
```

```
# Line Chart
```

```
plot(employee,  
      score,  
      type="o",  
      col="blue",  
      xlab="Employee",  
      ylab="Performance",  
      main="Employee Performance")
```

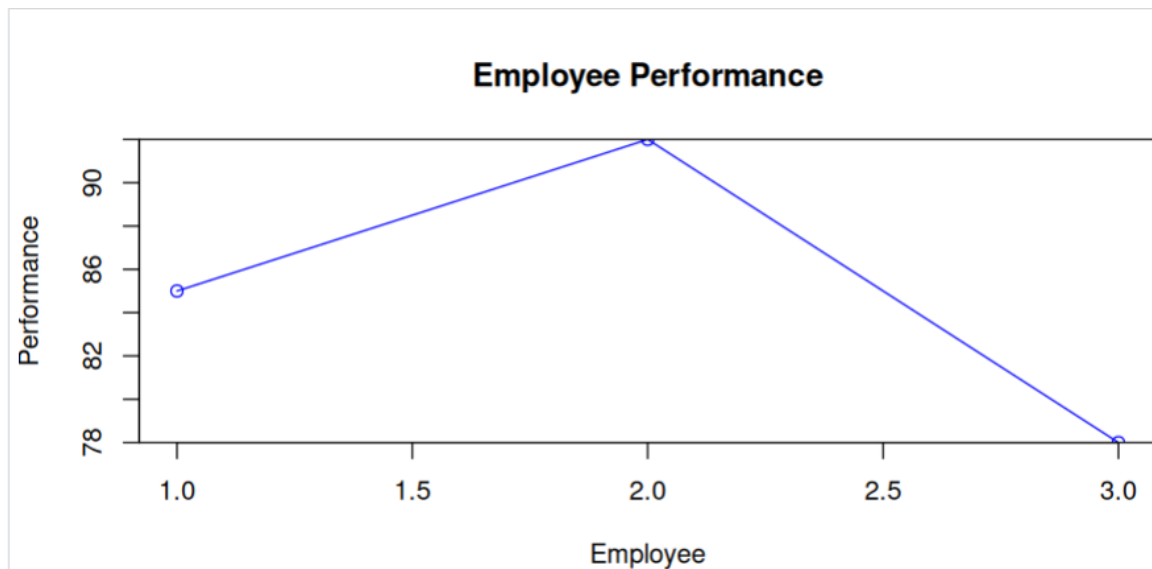
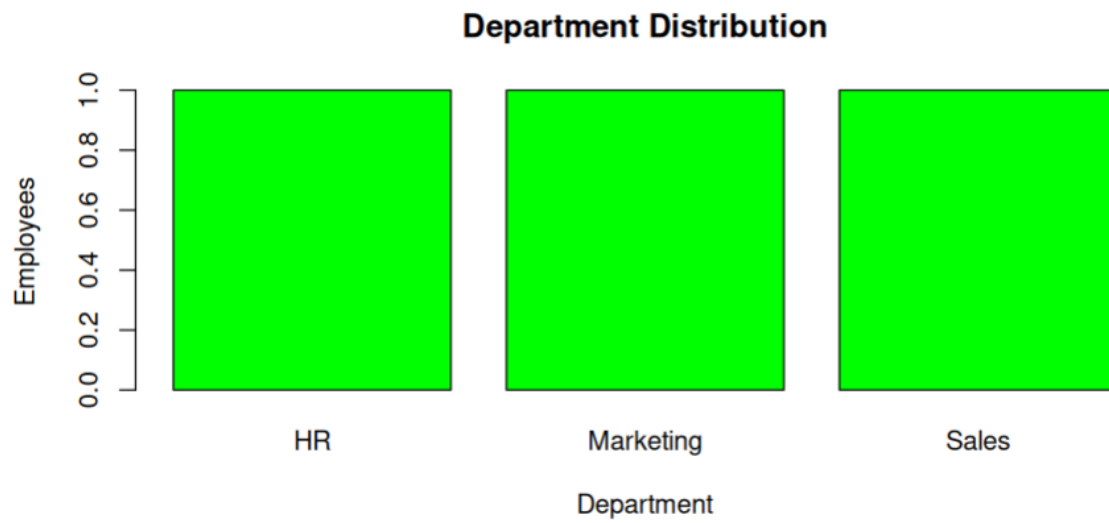
```
# Bar Chart
```

```
barplot(table(dept),  
        col="green",  
        main="Department Distribution",  
        xlab="Department",  
        ylab="Employees")
```

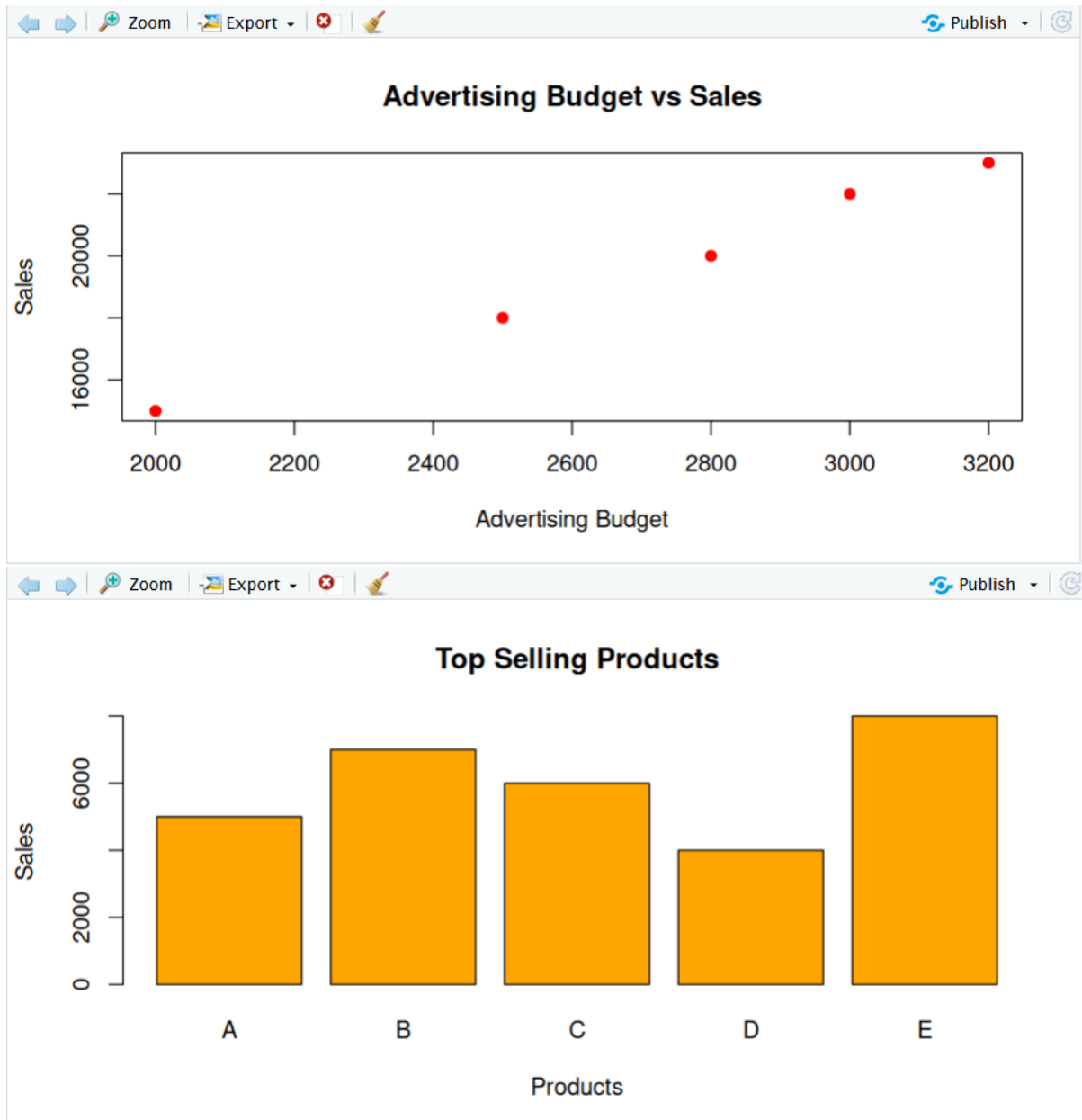
```
# Table
```

```
performance <- data.frame(employee,dept,service,score)  
print(performance)
```

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SET 9 – Online Learning Activity

Dataset

```
student <- c("L01","L02","L03","L04","L05","L06")
```

```
gender <- c("Male","Female","Male","Female","Male","Female")
```

```
course <- c("R","R","SQL","R","R","SQL")
```

```
study <- c(3.5,4.2,2.0,5.0,2.5,4.0)
```

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```
videos <- c(12,15,8,18,9,14)
```

```
quiz <- c(78,85,65,92,70,88)
```

```
login <- as.Date(c("2025-01-05","2025-01-05","2025-02-08",  
                  "2025-02-08","2025-03-12","2025-03-12"))
```

```
# Histogram
```

```
hist(quiz,  
     col="skyblue",  
     main="Quiz Score Distribution",  
     xlab="Quiz Score")
```

```
# Boxplot
```

```
boxplot(quiz~course,  
        col=c("orange","green"),  
        main="Quiz Score by Course",  
        xlab="Course",  
        ylab="Quiz Score")
```

```
# Scatter Plot
```

```
symbols(study,  
        quiz,  
        circles=videos,  
        inches=0.3,  
        bg=rgb(0,0,1,0.4),  
        xlab="Study Time",  
        ylab="Quiz Score",  
        main="Study Time vs Quiz Score")
```

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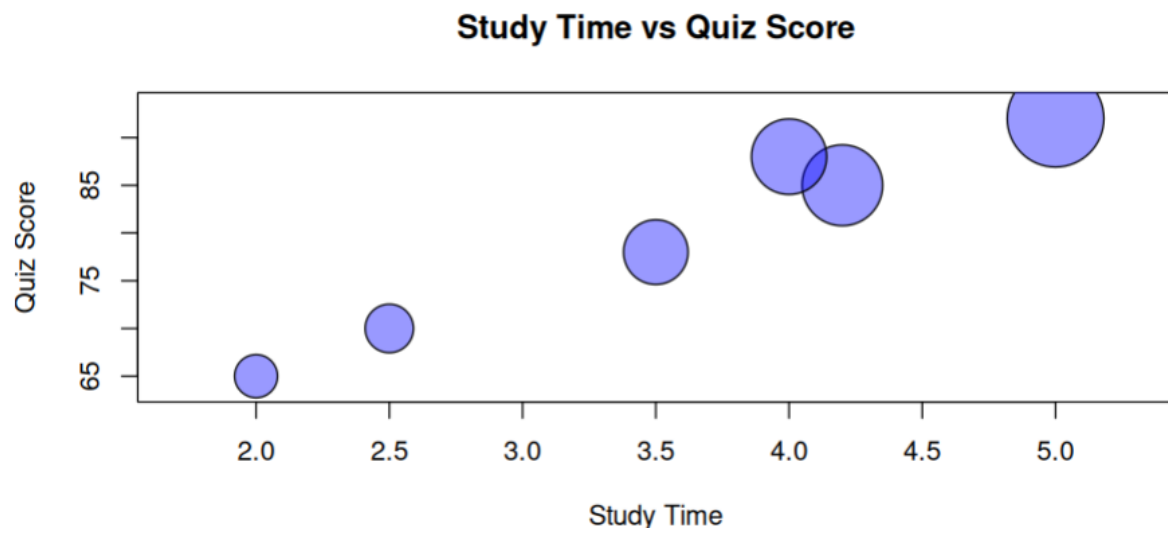
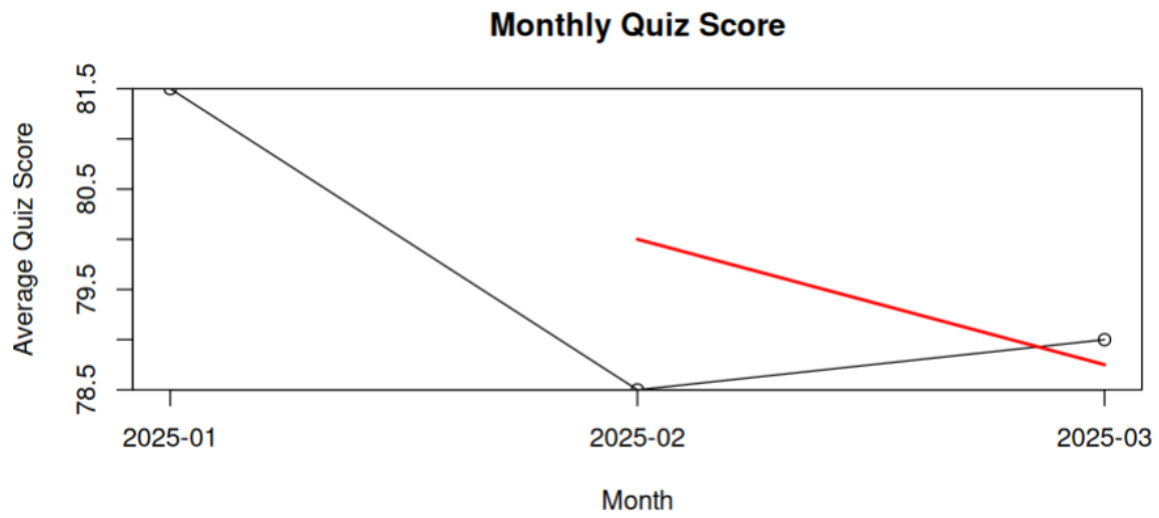
```
# Monthly Average
month <- format(login,"%Y-%m")
avg <- aggregate(quiz,
                  by=list(month),
                  FUN=mean)

print(avg)

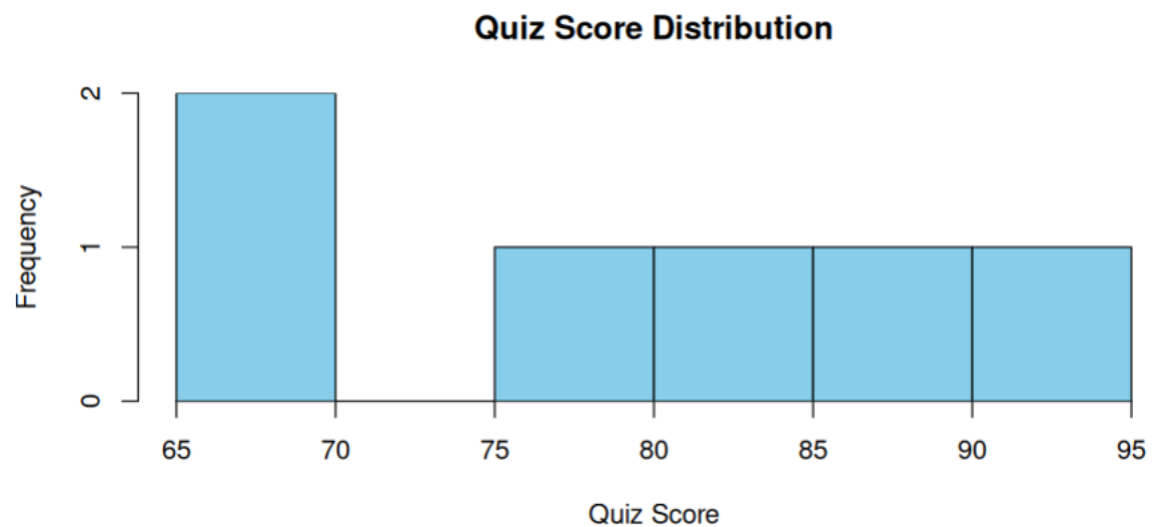
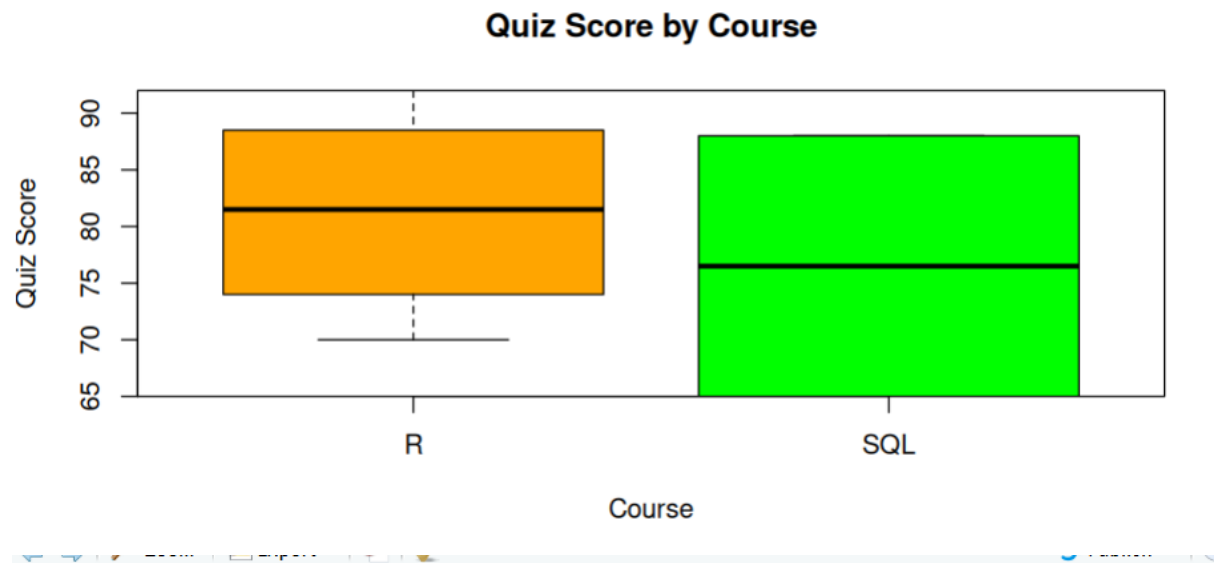
plot(avg$x,
      type="o",
      xaxt="n",
      xlab="Month",
      ylab="Average Quiz Score",
      main="Monthly Quiz Score")
axis(1,at=1:nrow(avg),labels=avg$Group.1)

# Moving Average
lines(stats::filter(avg$x,rep(1/2,2),sides=1),
      col="red",
      lwd=2)
```

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SET 10 – Survey Responses Analysis

```
survey <- 1:3
```

```
q1 <- c("A","B","C")
```

```
q2 <- c("B","A","A")
```

```
q3 <- c("C","D","B")
```

```
# Grouped Bar Chart
```

```
tab1 <- table(q1)
```

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```
barplot(tab1,  
        beside=TRUE,  
        col="cyan",  
        main="Question 1 Responses",  
        xlab="Responses",  
        ylab="Count")
```

Stacked Bar Chart

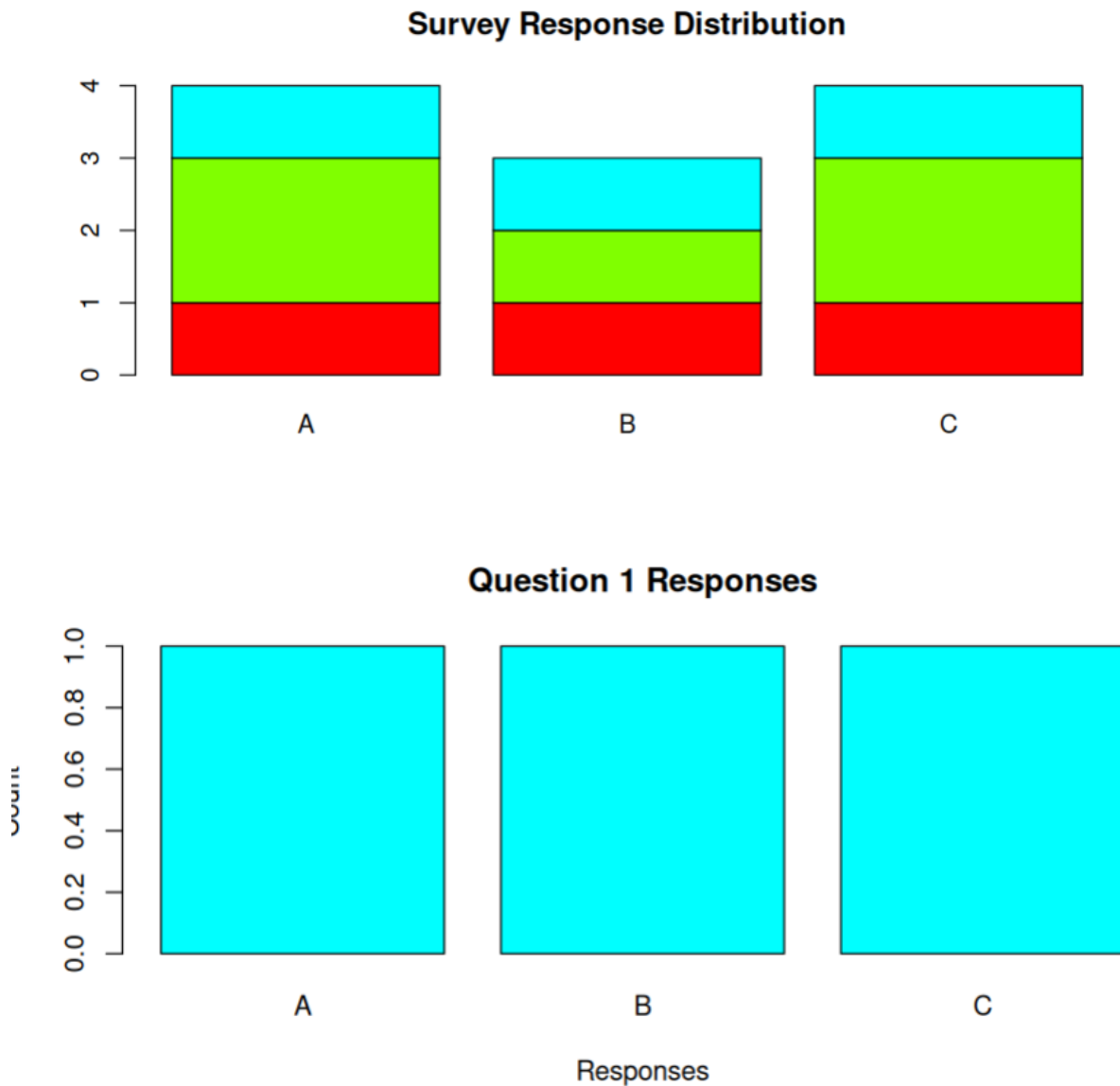
```
responses <- rbind(table(q1),  
                   table(q2),  
                   table(q3))
```

```
barplot(responses,  
        beside=FALSE,  
        col=rainbow(4),  
        legend=rownames(responses),  
        main="Survey Response Distribution")
```

Table

```
survey_data <- data.frame(survey,q1,q2,q3)  
print(survey_data)
```

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SET 11 – Product Category Analysis

Dataset

```
category <- c("Electronics","Clothing","Appliances")
```

```
sales <- c(50000,35000,40000)
```

1. Funnel Chart (Horizontal Bar Plot)

```
barplot(rev(sales),
```

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```
horiz=TRUE,  
names.arg=rev(category),  
col=c("red","blue","green"),  
main="Sales Conversion Funnel",  
xlab="Sales")
```

2. Table

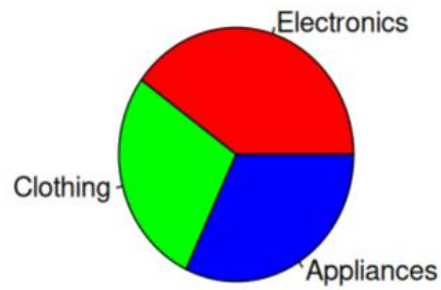
```
product_data <- data.frame(Category=category,  
                             Sales=sales)  
print(product_data)
```

4. Pie Chart

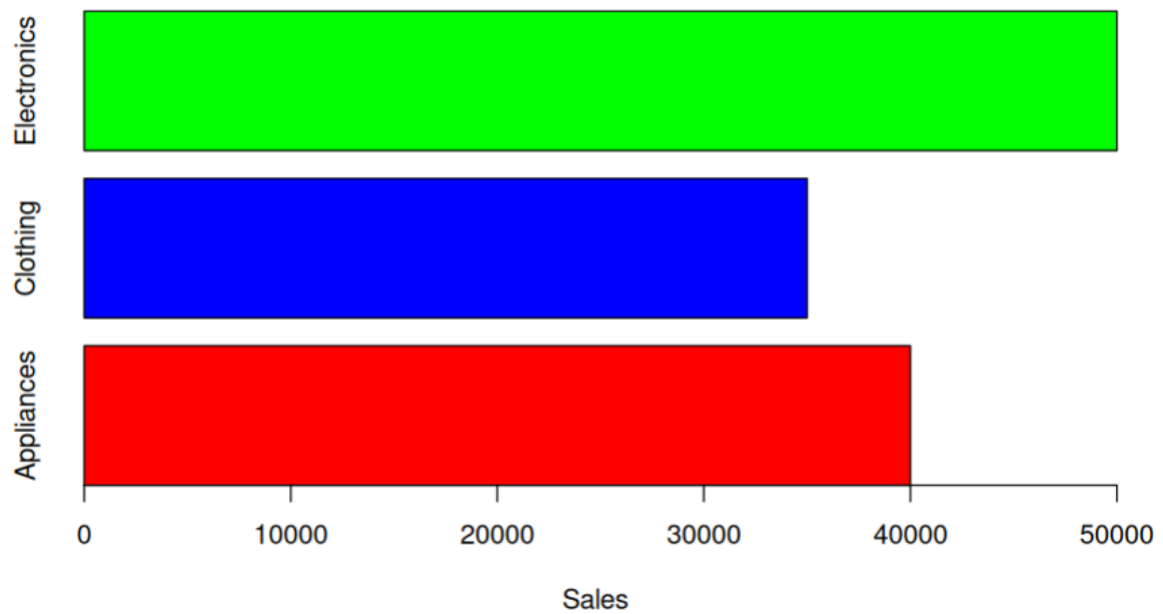
```
pie(sales,  
    labels=category,  
    col=rainbow(3),  
    main="Sales Distribution by Category")
```

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Sales Distribution by Category



Sales Conversion Funnel



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SET 12 – Website Traffic

Dataset

```
date <- as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03"))
```

```
views <- c(1500,1600,1400)
```

```
ctr <- c(2.3,2.7,2.0)
```

1. Line Chart

```
plot(date,  
      views,  
      type="o",  
      col="blue",  
      pch=19,  
      main="Daily Page Views",  
      xlab="Date",  
      ylab="Page Views")
```

2. Bar Chart

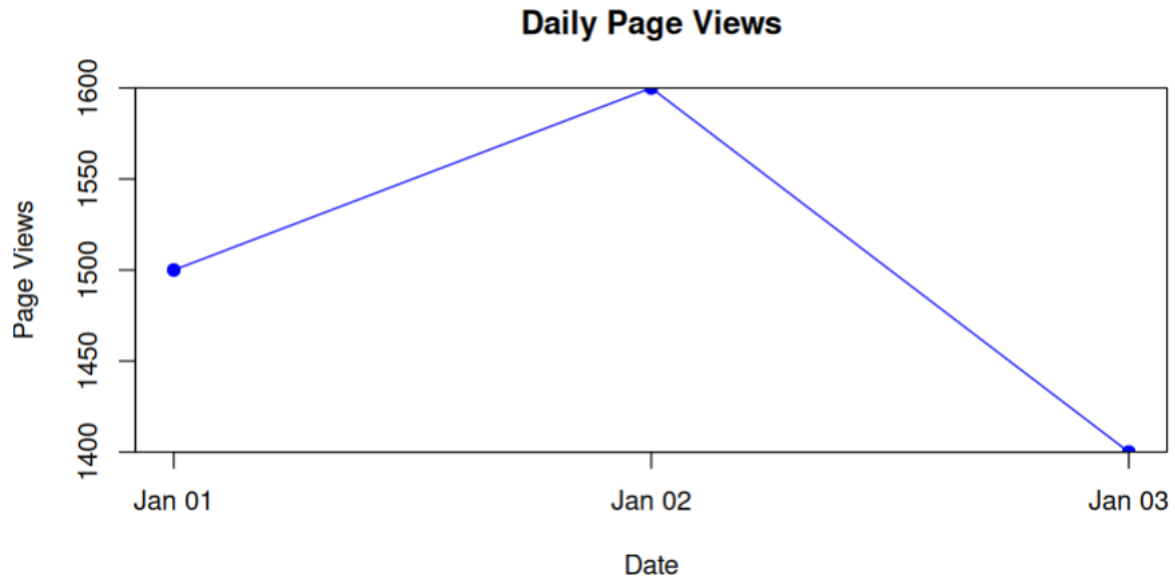
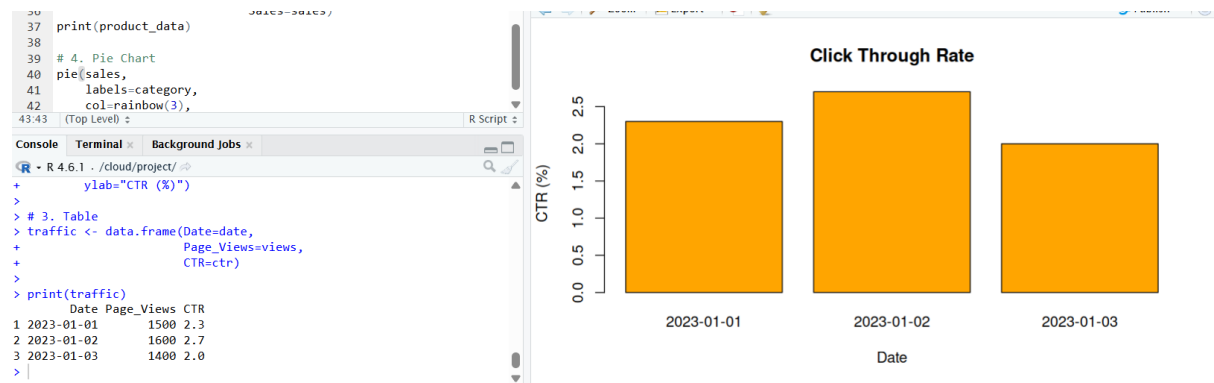
```
barplot(ctr,  
        names.arg=date,  
        col="orange",  
        main="Click Through Rate",  
        xlab="Date",  
        ylab="CTR (%)")
```

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3. Table

```
traffic <- data.frame(Date=date,  
                      Page_Views=views,  
                      CTR=ctr)
```

```
print(traffic)
```



SET 13 – Geographic Data

Dataset

```
city <- c("City A","City B","City C")
```

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```
population <- c(500000,700000,600000)
```

```
temperature <- c(75,68,80)
```

```
elevation <- c(1000,800,1200)
```

```
# 1. Map (Approximation)
```

```
plot(population,  
      elevation,  
      pch=19,  
      col="blue",  
      xlab="Population",  
      ylab="Elevation",  
      main="Geographic Distribution")  
text(population,elevation,labels=city,pos=3)
```

```
# 2. Scatter Plot
```

```
plot(temperature,  
      population,  
      pch=19,  
      col="red",  
      xlab="Average Temperature",  
      ylab="Population",  
      main="Temperature vs Population")
```

```
# Insight
```

```
cat("Population does not show a strong relationship with temperature.\n")
```

```
# 3. Table
```

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```
geo <- data.frame(city,population,temperature,elevation)
```

```
print(geo)
```

